Ordinal addition is not commutative

The Open Logic Project contributors

openlogic:set-theory/ord-arithmetic:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel ordsumnotcommute of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the recursion equations for ordinal addition at limit ordinals
- the strict-supremum description of ordinal addition at omega
- the successor of omega

Proposition 0.1. *Ordinal addition is not commutative;* $1 + \omega = \omega < \omega + 1$.

Proof. Note that $1 + \omega = \subsetneq_{n<\omega} (1 + n) = \omega \in \omega \cup \{\omega\} = +1\omega = \omega + 1$. □

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